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Def v(coalition):
    Return sum(player_scores[player] for player in coalition)

Import itertools
Import random
From collections import defaultdict

Players = list(player_scores.keys())
Num_samples = 10000 # You can increase this for better approximation

Shapley_values = defaultdict(float)

For _ in range(num_samples):
    Perm = random.sample(players, len(players))
    Current_coalition = set()
    Prev_value = 0

    For player in perm:
        Curr_value = v(current_coalition | {player})
        Marginal_contribution = curr_value - prev_value
        Shapley_values[player] += marginal_contribution
        Current_coalition.add(player)
        Prev_value = curr_value

# Average over number of samples
For player in players:
    Shapley_values[player] /= num_samples

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# Print result
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For player, value in shapley_values.items():
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    Print(f'{player}: Shapley Value  $\approx$  {value:.4f}')
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